

EXP NO 19: Implementation of Naive Bayes classification for Bank Loan prediction

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AIM

To implement Bank Loan Prediction using the Naive Bayes classification algorithm and predict whether a loan application will be approved or rejected based on applicant details.

PROCEDURE :

1. Collect the bank loan dataset containing applicant information such as gender, income, education, loan amount, credit history, and loan status.
2. Preprocess the dataset by handling missing values and converting categorical attributes into numerical values.
3. Split the dataset into training and testing data.
4. Train the Naive Bayes classifier using the training dataset and learn the relationship between applicant features and loan approval.
5. Predict loan approval for the test data and evaluate the model using accuracy and a confusion matrix.

CODE:

```
# Naive Bayes Classification for Bank Loan Prediction
```

```
import pandas as pd
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import LabelEncoder
```

```
from sklearn.naive_bayes import GaussianNB
```

```
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
# Load dataset
```

```
df = pd.read_csv("loan_data.csv")
```

```
# Fill missing values
```

```
df = df.fillna(method="ffill")

# Encode categorical columns
le = LabelEncoder()
for col in df.select_dtypes(include="object").columns:
    df[col] = le.fit_transform(df[col])

# Features and target
X = df.drop("Loan_Status", axis=1)
y = df["Loan_Status"]

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train Naive Bayes model
model = GaussianNB()
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Evaluation
print("Predicted:", y_pred)
print("Actual :", y_test.values)
print("Accuracy :", accuracy_score(y_test, y_pred))
print("Confusion Matrix:")
```

```
print(confusion_matrix(y_test, y_pred))
```

OUTPUT:

```
Predicted: [1 2 0 ... 0 1 0]
Actual    : [0 2 0 ... 1 2 0]
Accuracy  : 0.6423529411764706
Confusion Matrix:
[[968 291   1]
 [492 914  67]
 [ 69 296 302]]
```

Result:

The Naive Bayes classifier successfully classified bank loan applications with good prediction accuracy.